

Mark schemes

Q1.

(a) $20 \sin 50^\circ t - 4.9 t^2 = 0$

M1: Equation to find time, with $y = 0$,

$u = 20 \sin 50^\circ$ or $u = 20 \cos 50^\circ$ and ± 9.8 or $\pm g$.

A1: Correct equation

M1A1

$$t = \frac{20 \sin 50^\circ}{4.9} \text{ or } 3.126 \dots = 3.13 \text{ s} \quad \mathbf{AG}$$

dM1: Solving for t .

A1: Correct time from correct working.

Must see division by 4.9 oe or more than 3 sf

Verification methods can only gain first 2 marks

$$t = \frac{15.3}{4.9} = 3.12 \text{ or } 3.13$$
 Special case scores
 M1A1dM1A0

dM1A1

OR

$$0 = 20 \sin 50^\circ - 9.8t$$

$$t = \frac{20 \sin 50^\circ}{9.8} = 1.563$$

$$T = 2 \times 1.563$$

M2: doubling time to max height (could use cos instead of sin) but must use ± 9.8 or $\pm g$.

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$$= 3.13$$

A2: Correct time from correct working.

Don't penalise use of $g = 9.81$ if already done earlier on script. Would obtain time as 3.12 seconds.

Note: If using a memorised formula either 4 marks if final answer correct, 3 marks if substituted correctly, otherwise zero.

Special case

$$T = 2 \times 1.56 = 3.12 \text{ or } 3.13 \text{ scores M2A1}$$

(A2)

4

(b) $PQ = 20 \cos 50^\circ \times 3.127 = 40.2 \text{ m}$

M1: Calculation of range, could use sin instead of cos.

M1

*A1: Correct range
Accept 40.1*

A1

2

- (c) No change because a greater mass would not change the acceleration. OR Mass is not used in the equations.

B1: No change

B1

B1: Explanation following a correct statement.

B1

2

- (d) $0 = (20 \sin 50^\circ)^2 + 2 \times (-9.8)s$

*M1: Equation to find height, with
 $u = 20 \sin 50^\circ$ or $u = 20 \cos 50^\circ$ and ± 9.8
or $\pm g$ (and t between 1.56 and 1.57 if
method 2 used).*

A1: Correct equation

M1A1

$$s = \frac{(20 \sin 50^\circ)^2}{2 \times 9.8} = 12.0 \text{ m}$$

*A1: Correct height. Accept 12 or 11.9 or
AWRT 12.0*

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A1

OR

$$t = \frac{3.13}{2} = 1.565$$

In method 2, no marks awarded for just finding t .

$$h = 20 \sin 50^\circ \times 1.565 - 4.9 \times 1.565^2$$

(M1)(A1)

$$= 12.0$$

(A1)

*Don't penalise use of $g = 9.81$ if already
done earlier on script. Should still get 12.*

Note: If using a memorised formula either 3 marks if final answer correct, 2 marks if substituted correctly, otherwise zero.

3

- (e) 20 m s^{-1} at 50° below the horizontal.

B1: Speed AWRT 20

B1

B1: Direction AWRT 50° . Must indicate below, or down. Could be implied by a diagram.

B1

2

[13]

Q2.

- (a) (i) $20 \times 9.8 = R + 60 \sin 30^\circ$

M1: Equation or expression for normal reaction with mg or $20g$ or 196 and $60 \sin 30^\circ$ or $60 \cos 30^\circ$.

A1: Correct equation or expression with correct signs.

M1A1

$(R =) 20 \times 9.8 - 60 \sin 30^\circ = 166 \text{ N}$ AG

A1: Correct value from correct working. Must be positive.

Don't penalise use of $g = 9.81$ if already done earlier on script. Should still get 166, but from 166.2.

A1

3

- (ii) $166\mu = 60 \cos 30^\circ$

M1: Use of $F = \mu R$, with $R = 166$ or 166.2 . Do not allow inequalities here.

M1: Resolving horizontally with $\cos 30^\circ$ or $\sin 30^\circ$ oe

A1: Correct equation

Examples:

$166\mu = 60$ M1M0A0

$166\mu = -60 \cos 30^\circ$ M1M1A0

M1M1A1

$$\mu = \frac{60 \cos 30^\circ}{166}$$

$$= 0.313$$

A1: Correct coefficient of friction.

A1

4

(b) $20 \times 0.8 = T \cos 30^\circ - 0.313(20 \times 9.8 - T \sin 30^\circ)$

B1: $20g - T \sin 30^\circ$ oe seen.

*M1: Three term equation of motion,
where normal reaction is dependent on T .*

A1F: Correct equation

B1M1A1F

$$T = \frac{20 \times 0.8 + 0.313 \times 20 \times 9.8}{\cos 30^\circ + 0.313 \sin 30^\circ} = 75.6 \text{ N}$$

dm1: Solving for T including factorisation.

A1F: Correct tension.

AWRT 75.6

*Follow through incorrect values of μ
from part (a).*

*Don't penalise use of $g = 9.81$ if already
done earlier on script. Should get 75.7.*

Allow 75.8 if intermediate values rounded.

5

[12]

Q3.

(a) $-3 = 70 \sin 10^\circ t - 4.9t^2$

Vertical equation to find t containing a 3

M1

Correct RHS of equation

A1

$$4.9t^2 - 70 \sin 10^\circ t - 3 = 0$$

Correct LHS of equation

A1

$$t = \frac{70 \sin 10^\circ \pm \sqrt{(70 \sin 10^\circ)^2 - 4 \times 4.9 \times (-3)}}{2 \times 4.9}$$

Solving for t

dM1

$$t = -0.226 \text{ or } 2.71$$

Selecting the positive t

dM1

$$t = 2.71$$

Correct time from correct working

A1

6

(b) $s = 70 \cos 10^\circ \times 2.71 = 187$

Finding horizontal distance

M1

Correct distance

A1

2

(c) $v_x = 70 \cos 10^\circ = 68.9$

Calculating horizontal component of velocity

M1

Correct value

A1

2

(d) $v_y = 70 \sin 10^\circ - 9.8 \times 2.71 = -14.4$

Calculating vertical component of velocity at $t = 2.71$

M1

Correct value

A1

$$v = \sqrt{68.9^2 + 14.4^2} = 70.4$$

Finding speed

dM1

Correct speed

A1

4

[14]

Q4.

(a) (i) $0.2a = -0.2 \times 9.8 \sin 20^\circ$

AG

$$a = -9.8 \sin 20^\circ = -3.35 \text{ m s}^{-2}$$

Two term equation of motion with weight resolved

M1

Correct equation

A1

Correct acceleration from correct working

A1

SC No negative sign but otherwise correct award M1A1A0

Allow $a = -g \sin 20^\circ$

3

(ii) $0 = 4^2 + 2 \times (-3.35)s$

Use of constant acceleration equation with $v = 0$ and $u = 4$

M1

Correct equation

A1

$$s = \frac{16}{6.7} = 2.39 \text{ m}$$

Correct distance

A1

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3

- (iii) The puck slides back down the slope as the puck is at rest and the resultant force is now acting down the slope / no friction / smooth slope.

Slides back down

B1

Acceptable explanation

E1

2

(b) (i) $R = 0.2 \times 9.8 \cos 20^\circ$

AG

Finding normal reaction by resolving. Must see a trig term.

M1

$$F = 0.5 \times 0.2 \times 9.8 \cos 20^\circ$$

Use of $F = \mu R$

M1

$$= 0.921 \text{ N}$$

Correct friction from correct working.

A1

3

(ii) $0.2a = -0.921 - 0.2 \times 9.8 \sin 20^\circ$

Three term equation of motion with the weight resolved

M1

Correct equation

A1

$$a = -7.96 \text{ m s}^{-2}$$

Correct acceleration (with or without the minus sign, applied to both A1 marks)

A1

3

- (iii) The puck stays at rest because the friction has a maximum of 0.921 and the component of the weight down the slope is less (0.670)

Stays at rest

B1

Acceptable explanation

dE1

2

[16]

Q5.

- (a) It is a particle /No air resistance / lift forces act on the ball.

Particle

B1

*Other acceptable assumption
Deduct one mark for each additional incorrect assumption.*

B1

2

(b) $V \sin 40^\circ t - \frac{1}{2} \times 9.8 t^2 = 0$

Vertical equation to find t.

M1

*Correct equation
(Equals zero may be implied)*

A1

$$t = \frac{V \sin 40^\circ}{4.9}$$

Solving for t

dM1

Correct t

A1

$$s = V \cos 40^\circ \times \frac{V \sin 40^\circ}{4.9}$$

AG

$$= \frac{V^2 \cos 40^\circ \sin 40^\circ}{4.9}$$

Finding range with their t

M1

*Correct range from correct working
SC Quoting the formula for the range 2 marks.*

A1

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6

(c) $76 < \frac{V^2 \cos 40^\circ \sin 40^\circ}{4.9} < 82$

$$\sqrt{\frac{76 \times 4.9}{\cos 40^\circ \sin 40^\circ}} < V < \sqrt{\frac{82 \times 4.9}{\cos 40^\circ \sin 40^\circ}}$$

$$27.5 < V < 28.6$$

An equation to find one value of V.

M1

Correct value for V

A1

Other value of V correct

A1

Correct range of values
 Accept 27.5 – 28.6 but not 28.6-27.5
 For using values close to 76 and 82
 deduct one mark.

A1

4

[12]

Q6.

(a) $0 = 16 \sin \theta \times 2 - 4.9 \times 2^2 = 0$ **AG**

Equation to find θ based on height being zero.

M1

Correct equation

A1

$$\sin \theta = \frac{19.6}{32}$$

$$\theta = 37.8^\circ$$

Correct angle from correct working

A1

3

(b) $2 = 16 \sin 37.8t - 4.9t^2$

Equation for height of 2

M1

Correct equation

A1

$$4.9t^2 - 16 \sin 37.8t + 2 = 0$$

$$t = \frac{16 \sin 37.8 \pm \sqrt{(16 \sin 37.8)^2 - 4 \times 2 \times 4.9}}{2 \times 4.9}$$

Solving quadratic equation

m1

$$t = 1.77 \text{ or } t = 0.23$$

Correct solutions

A1

Greater than 2 metres for 1.54 seconds

Difference between solutions

A1

5

[8]

Q7.

(a) (i) $0 = 40 \sin 35^\circ t - 4.9t^2$

Equation to find time of flight with 40,

sin/cos35° and -4.9 or $-\frac{g}{2}$

M1

Correct equation

A1

$$t = \frac{40 \sin 35^\circ}{4.9} = 4.68 \text{ s}$$

AG

Solving for t

m1

*Correct time from correct working
 Note: candidates must have a method for the complete time of flight before any marks can be awarded.
 Condone the use of a formula for the time of flight.*

A1

4

(ii) $AB = 40 \cos 35^\circ \times 4.682 = 153 \text{ m}$

Calculating the range using 40, cos/sin35° and 4.68 and acceleration zero.

M1

*Correct range
 Accept AWRT 153*

A1

2

(b) $-1 = 40 \sin 35^\circ t - 4.9t^2$

$$4.9t^2 - 40 \sin 35^\circ t - 1 = 0$$

Equation to find time of flight with a ± 1 ,

40, sin/cos35° and -4.9 or $-\frac{g}{2}$

M1

Correct terms

Correct signs

A1

A1

$$t = \frac{40 \sin 35^\circ \pm \sqrt{(40 \sin 35^\circ)^2 - 4 \times 4.9 \times (-1)}}{2 \times 4.9}$$

Solving quadratic equation

m1

$$t = 4.73 \text{ or } t = -0.0432$$

Accept AWRT 4.73 or 4.72

A1

$$t = 4.73$$

*Rejection of negative solution indicated
(Only 4.73 or 4.72 given award 5/6 marks)*

A1

Alternative methods based on finding two times.

For example,

$$t = 4.682 + 0.044 = 4.73$$

$$t = 2.341 + 2.384 = 4.73$$

Addition of two times

(M1)

Use of AWRT 4.68 or AWRT 2.34

(A1)

Calculation of time for 'second' part

(m1)

Correct expression for time for 'second' part

(A1)

Correct time (Allow AWRT 0.04 or AWRT 2.38)

(A1)

Correct total time

Accept 4.72

(A1)

6

[12]

Q8.

(a) $5 = 10 \cos \alpha.t$

M1

$$t = \frac{5}{10 \cos \alpha}$$

A1

$$1 = -\frac{1}{2}(9.8)t^2 + 10 \sin \alpha t$$

M1A1

$$1 = -\frac{1}{2}(9.8)\frac{25}{100 \cos^2 \alpha} + 10 \sin \alpha \frac{5}{10 \cos \alpha}$$

Dependent on both M1s

M1

$$1 = -\frac{1}{2}(9.8)\frac{25}{100}(1 + \tan^2 \alpha) + 10 \sin \alpha \frac{5}{10 \cos \alpha}$$

A1

$$49 \tan^2 \alpha - 200 \tan \alpha + 89 = 0$$

Answer given

A1

7

$$(b) \quad \tan \alpha = \frac{200 \pm \sqrt{40000 - 4(49)(89)}}{2 \times 49}$$

M1

$$= 3.57, 0.508$$

AWRT

A1

$$\alpha = 74.4^\circ, 26.9^\circ$$

A1F

3

- (c) (i) $10 \cos 26.9^\circ = 8.92$ (or 8.91) > 8
 $\Rightarrow$ The can will be knocked off the wall

Both values checked

M1

$$10 \cos 74.4^\circ = 2.69 < 8$$

*Acc. of both results
Correct conclusions*

A1F

$\Rightarrow$ The can will not be knocked off the wall

E1

ALTERNATIVE

The can will be knocked off the wall if

$$10 \cos \alpha > 8$$

$$\cos \alpha > 0.8$$

$$\alpha < 36.9^\circ$$

M1A1

So, for $\alpha = 26.9^\circ$ the can will be knocked off

and for $\alpha = 74.4^\circ$, the can will not be knocked off E1

3

(ii) $x = ut$

$$t = \frac{5}{10 \cos 26.9^\circ}$$

$$v = 10 \sin 26.9^\circ - 9.8 \left(\frac{5}{10 \cos 26.9^\circ} \right)$$

Any correct use of equations

M1

$$v = -0.970$$

A1F

$$\tan \theta = \frac{-0.970}{8.92}$$

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M1

$$\theta = -6.2^\circ$$

At an angle of depression of 6.2°

AWRT 6°

A1F

4

[17]

Q9.

(a) $6 = 20 \sin 60^\circ t - 4.9t^2$

Vertical equation including a 6

M1

Correct terms

A1

$$4.9t^2 - 20 \sin 60^\circ t + 6 = 0$$

Correct signs

A1

$$t = \frac{20 \sin 60^\circ \pm \sqrt{(20 \sin 60^\circ)^2 - 4 \times 4.9 \times 6}}{2 \times 4.9}$$

Solving quadratic equation

dM1

$$t = 0.389 \text{ or } t = 3.15$$

$$t = 0.389 \text{ s}$$

Correct solution selected from correct working; AG

A1

5

(b) $x = 20 \cos 60^\circ \times 0.389 = 3.89 \text{ m}$

Finding distance. Correct distance

M1A1

2

(c) $v_x = 20 \cos 60^\circ = 10$

Correct horizontal component

B1

$$v_y = 20 \sin 60^\circ - 9.8 \times 0.389 = 13.5$$

Finding vertical component of velocity

M1

Correct vertical component

A1

$$v = \sqrt{10^2 + 13.5^2} = 16.8 \text{ m s}^{-1}$$

Finding magnitude. Correct magnitude

dM1A1

5

[12]

Q10.

(a) $F = \sqrt{6^2 + 5^2}$

Obtaining an equation for F with square or root. Correct equation

M1A1

$$= \sqrt{61} = 7.81$$

Correct force

A1

Alt

$$\alpha = \tan^{-1}\left(\frac{5}{6}\right) = 39.8^\circ$$

$$F = \frac{6}{\cos 39.8} = 7.81 \text{ or}$$

Equation for F with a value for α . Correct equation

(M1A1)

$$F = \frac{5}{\sin 39.8} = 7.81$$

Correct force

(A1)

3

(b) $\alpha = \tan^{-1}\left(\frac{5}{6}\right) \text{ or } \cos^{-1}\left(\frac{6}{7.81}\right) \text{ or } \sin^{-1}\left(\frac{6}{7.81}\right)$

Obtaining an equation for α using trigonometry. Correct equation (using their F)

M1A1

$$= 39.8^\circ$$

Correct angle

A1

Accept values between 39.7 and 39.9

Alt

$$\frac{\sin \alpha}{5} = \frac{\sin 90^\circ}{\sqrt{61}}$$

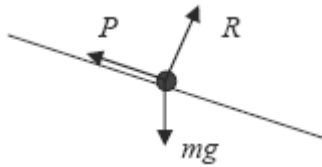
$$\alpha = 39.8^\circ$$

3

[6]

Q11.

(a)



Correct diagram with arrows and labels

Must not use F instead of P

Condone resistance instead of P

B1

1

(b) $P = 100 \times 9.8 \sin 4^\circ$

Resolving weight (must see 100)

M1

Using $\sin 4^\circ$ or $\cos 86^\circ$

M1

$= 68.4$

AG *Correct P from correct working*

A1

3

(c) $100a = 100 \times 9.8 \sin 5^\circ - 100 \times 9.8 \sin 4^\circ$

Three term equation of motion

M1

Weight resolved correctly

A1

Correct equation

A1

$$a = \frac{100 \times 9.8 \sin 5^\circ - 100 \times 9.8 \sin 4^\circ}{100}$$

$= 0.171$

Correct a . (Accept 0.170 or 0.17)

A1

4

(d) You would expect P to vary with the speed of the car.

Correct explanation

B1

Q12.

(a) $0^2 = (50 \sin 40^\circ)^2 + 2 \times (-9.8)h$

Equation for h with $v = 0$ and a component of velocity. Correct equation

M1A1

$$h = \frac{(50 \sin 40^\circ)^2}{2 \times 9.8} = 52.7$$

Solving for h

dM1

Correct h

A1

Alt

$$0 = 50 \sin 40^\circ - 9.8t$$

Equation for t with $v = 0$ and a component of velocity

(M1)

$$t = \frac{50 \sin 40^\circ}{9.8} = 3.280$$

Correct t

(A1)

$$h = 50 \sin 40^\circ \times 3.280 - \frac{1}{2} \times 9.8 \times 3.280^2$$

Expression for h with a component of velocity

(dM1)

$$= 52.7$$

ALLOW 52.6

Correct h

(A1)

(b) $6 = 50 \sin 40^\circ t - 4.9t^2$

Forming a quadratic in t . Correct terms

with any signs

M1A1

$$0 = 4.9t^2 - 50 \sin 40^\circ t + 6$$

Correct equation

A1

$$t = \frac{50 \sin 40^\circ \pm \sqrt{(50 \sin 40^\circ)^2 - 4 \times 4.9 \times 6}}{2 \times 4.9}$$

Solving quadratic

dM1

$$= 0.192 \text{ or } 6.37$$

$$t = 6.37$$

Correct solution selected

A2

Alt

$$46.7 = 4.9 t_1^2$$

Finding two times

(M1)

Equation for time to go down

(dM1)

$$t_1 = 3.087$$

Correct time

(A1)

$$t_2 = 3.280$$

Time to go up

(A1)

$$t = 3.087 + 3.280 = 6.37$$

Correct total

(A2)

6

[10]

Q13.

(a) $2.45 = \frac{1}{2} \times 9.8t^2$

Equation for time to ground

M1

Correct equation

A1

$$t = \sqrt{\frac{2.45}{4.9}} = 0.707 \text{ seconds (to 3 sf)}$$

AG Correct time from correct working

A1

3

(b) $s = 20 \times 0.707 = 14.1 \text{ m (to 3sf)}$

Using $20 \times$ time from part (a)

M1

Correct distance

A1

2

(c) $v_y = 0.707 \times 9.8 = 6.929$

Finding vertical component of velocity

M1

Correct vertical component

A1

$$\theta = \tan^{-1}\left(\frac{6.929}{20}\right) = 19.1^\circ$$

Using tan to find the angle

dM1

Correct angle

A1

4

[9]

Q14.

(a) $T_1 \sin 35^\circ = T_2 \sin 35^\circ$

Resolving two forces and forming an equation, with different tensions for each string

M1

$$T_1 = T_2$$

Correct result from correct working

A1

OR

$$T_1 \cos 55^\circ = T_2 \cos 55^\circ$$

$$T_1 = T_2$$

2

(b) $T_1 \cos 35^\circ + T_2 \cos 35^\circ = 2 \times 9.8$

$$T_1 \cos 35^\circ + T_1 \cos 35^\circ = 2 \times 9.8$$

Resolving forces to form a three term vertical equation

M1

Correct equation

A1

T_1 or T_2 eliminated correctly

A1

$$T_1 = \frac{2 \times 9.8}{2 \cos 35^\circ} = 12.0 \text{ N (to 3sf)}$$

Solving for T_1 or T_2

dM1

*Correct tension
Accept 12 N or 11.9 N*

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(c) $2 \times 40 \cos 35^\circ = 9.8m$

Forming an equation with two tensions to find m

M1

Correct equation

A1

$$m = \frac{80 \cos 35^\circ}{9.8} = 6.69 \text{ kg}$$

*Correct mass
Accept 6.68*

A1

OR

$$m = \frac{40}{11.96} \times 2$$

(M1)(A1)

$$= 6.69 \text{ kg}$$

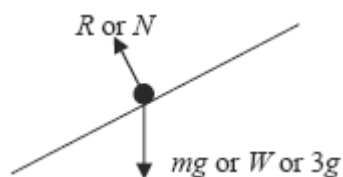
(A1)

3

[10]

Q15.

(a) (i)



Correct diagram with arrows and labels

B1

1

(ii) $3a = 3g \sin 30^\circ$

Two term equation of motion

M1

$$a = g \sin 30^\circ = 4.9 \text{ m s}^{-2}$$

AG Correct acceleration from correct working (Allow $a = g \sin 30^\circ$)

A1

2

(b) (i) $5 = \frac{1}{2}a \times 2^2$

Constant acceleration equation with $u = 0$

M1

$$a = 2.5 \text{ m s}^{-2}$$

*AG Correct answer from correct working.
(Use of $v = 5$ must be justified)*

A1

2

(ii) $3 \times 2.5 = 3g \sin 30^\circ - F$

Three term equation of motion

M1

Correct equation

A1

$$F = 3g \sin 30^\circ - 7.5$$

$$= 7.20 \text{ N (to 3 sf)}$$

Correct F

Accept 7.2 N

A1

3

(iii) $R = 3g \cos 30^\circ (= 25.46)$

Resolving perpendicular to the slope to find R

M1

Correct R

A1

$$7.2 = \mu \times 3g \cos 30^\circ$$

Use of $F = \mu R$

M1

Correct expression

A1F

$$\mu = \frac{7.2}{3g \cos 30^\circ} = 0.283$$

Correct μ

Accept 0.282

(Follow through from incorrect F from above, but not an incorrect R)

A1F

5

- (iv) Reduce a , as the air resistance would reduce the magnitude of the resultant force or because the air resistance increases as the velocity increases towards its terminal value

Reduces

B1

Explanation

Second B1 dependent on the first B1 mark

B1

Q16.

- (a) A particle or no spin

First assumption

B1

No air resistance or no wind or only gravity acting

Second assumption

B1

If more than 2 assumptions given, subtract one mark for each incorrect additional assumption

2

- (b) $0 = 25 \sin 40^\circ t - 4.9t^2$

Equation for time of flight

M1

Correct equation

A1

$$0 = t (25 \sin 40^\circ - 4.9t)$$

Solving for t

dM1

$$t = 0 \text{ or } t = \frac{25 \sin 40^\circ}{4.9}$$

$$\text{Time of flight} = 3.28 \text{ s}$$

*AG Correct final answer from correct working
(Verification method M1A1M1A0)*

A1

4

- (c) $s = 3.28 \times 25 \cos 40^\circ = 62.8 \text{ m}$

Finding range

M1

Correct range

A1

2

- (d) 25 m s^{-1} at 40° **below** the horizontal

Speed

B1

Direction

B1

2

(e) $v_{\min} = 25 \cos 40^\circ = 19.2 \text{ m s}^{-1}$

OR

$$v_{\min} = \frac{62.807}{3.2795} = 19.2 \text{ m s}^{-1}$$

Horizontal component of velocity

M1

*Correct speed
Accept 19.1 m s⁻¹*

A1

2

[12]

Q17.

(a) $y = ut \sin \alpha - \frac{1}{2} gt^2$

M1A1

$$x = ut \cos \alpha$$

M1

$$t = \frac{x}{u \cos \alpha}$$

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$$y = u \left(\frac{x}{u \cos \alpha} \right) \sin \alpha - \frac{1}{2} g \left(\frac{x}{u \cos \alpha} \right)^2$$

M1

$$y = x \tan \alpha - \frac{gx^2}{u^2 \cos^2 \alpha}$$

$$y = x \tan \alpha - \frac{gx^2}{2u^2} (1 + \tan^2 \alpha)$$

Answer given

A1

6

(b) (i) $1 = R \tan \alpha - \frac{10R^2}{2u^2} (1 + \tan^2 \alpha)$

M1

$$5R^2 \tan^2 \alpha - u^2 R \tan \alpha + 5R^2 + u^2 = 0$$

Answer given

A1

2

(ii) For real solutions of the quadratic:

$$u^4 R^2 - 20R^2(5R^2 + u^2) \geq 0$$

M1

$$R^2 \leq \frac{u^4 - 20u^2}{100}$$

$$R^2 \leq \frac{u^2(u^2 - 20)}{100}$$

Answer given

A1

2

(iii) $5^2 \leq \frac{u^2(u^2 - 20)}{100}$

$$u^4 - 20u^2 - 2500 \geq 0$$

Condone equation

M1

$$u_{\min}^2 = 61.0 \quad (\text{or } 10 + \sqrt{2600})$$

A1

$$u_{\min} = 7.81$$

3 sf required

A1F

3

[13]

Q18.

(a) $s = ut + \frac{1}{2} at^2$

$$0 = 2\frac{1}{2}ut - \frac{1}{2}gt^2$$

M1

*Full method required for time
(equations of motion, or standard result)*

A1

$$0 = t \left(2\frac{1}{2}u - \frac{1}{2}gt \right)$$

m1

$$t = \frac{5u}{g}$$

(if g = 9.8 used, lose last A1)

A1

4

(b) $OA = 6u \times \frac{5u}{g}$

M1

$$= \frac{30u^2}{g}$$

CAO

A1

2

(c) $\text{speed}^2 = (6u)^2 + \left(2\frac{1}{2}u \right)^2$

M1

$$\text{speed} = 6\frac{1}{2}u$$

CAO

A1

2

(d) Least speed, at top, = $6u$

B1

1

[9]

Q19.

(a) (i) $R = 80\cos 25^\circ$
component attempted

M1

correct component

A1

$R = 72.5\text{N}$
 CAO

A1

3

(ii) $F = 0.32 \times 72.5$
condone inequality

M1

$F = 23.2\text{N}$
 CAO

A1

2

(iii) $T + F = 80\cos 65^\circ$
3 forces direction correct, component

M2

attempted

A1

$T = 10.6\text{N}$
ft friction

A1ft

4

(iv) $T = F + 80\cos 65^\circ$
3 forces, direction correct, component

M1

attempted

A1

$T = 57.0\text{N} (57\text{N})$
ft friction

A1ft

3

$$(iv) \text{ Mass} = \frac{80}{g} = (8.16\text{kg})$$

B1

1

(b) $80 \cos 65^\circ - F = \text{mass} \times \text{acceleration}$
3 terms, component attempted

M1

$$10.6 = \frac{80}{g} \times \text{acc}$$

all correct

A1

$$\text{acc} = 1.30 \text{ ms}^{-2} (1.3 \text{ ms}^{-2})$$

CAO

A1

3

[16]

Q20.

(a) $57 = 24 \cos 40^\circ \times t$
component attempted and acceleration = 0

M1

all correct

A1

$$t = 3.10 \text{ sec}$$

CAO

A1

3

(b) $h = 24 \sin 40^\circ \times 3.1 - \frac{1}{2} \times 9.8 \times 3.1^2$
component attempted & acceleration = 9.8

M1

all correct

A1

$$h = 0.734 \text{ m}$$

ft one slip e.g. + 9.8 used

Accept 2 SF answer, AWR 0.71-0.74

A1F

3

(c) (i) horizontal, $u = 24 \cos 40^\circ = 18.39 \text{ ms}^{-1}$

Seen anywhere in (c), accept 18.4

B1

vertical, $v = 24 \sin 40^\circ - 9.8 \times 3.1$

component attempted & acceleration = 9.8

M1

$$v = -14.95 \text{ ms}^{-1}$$

(accept -15.0)

A1

$$V = \sqrt{(18.39^2 + (-14.95)^2)}$$

Use of candidate's u and 'new' v

M1

$$V = 23.7 \text{ ms}^{-1}$$

ft use of candidate's u and v

A1F

5

(ii) $\tan \theta = \frac{14.95}{18.39}$

Use of candidate's u and v (new)

Accept inverted ratio

M1

$$\theta = 39.1^\circ \text{ or } 39.2^\circ, 140.8^\circ, 140.9^\circ \text{ accept } \pm$$

ft use of candidate's u and v and V

A1

2

[13]

Q21.

(a) $y = -\frac{1}{2}gt^2 + 20\sin 30t$

M1A1

$$x = 20\cos 30t$$

M1

$$t = \frac{x}{20\cos 30}$$

A1

$$y = -\frac{1}{2}g\frac{x^2}{400\cos^2 30} + 20\sin 30\frac{x}{20\cos 30}$$

M1

$$y = x\tan 30 - \frac{gx^2}{800\cos^2 30}$$

AG

A1

6

(b) $2.5 = x\tan 30 - \frac{9.8x^2}{800\cos^2 30}$

$$9.8x^2 - 346x + 1500 = 0$$

substituting and tidying up

M1A1

$$x = \frac{346 \pm \sqrt{119716 - 58800}}{19.6}$$

M1

=30.3 (or 30.2) & 5.06 (or 5.05)
 answer: 30.3m (or 30.2m)
 at least 3 s.f. required

A1F

4

(c) no air resistance,

B1

the ball is a particle

B1

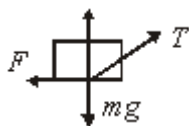
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etc.

[12]

Q22.

(a)



Correct diagram

B1

1

(b) $40 \cos 30^\circ - F = 25 \times 0.1$

$F = 40 \cos 30^\circ - 2.5 = 32.1 \text{ N}$

Three term equation of motion

Correct equation

M1

AG; correct force from correct working

A1

A1

3

(c) $R + 40 \sin 30^\circ = 25 \times 9.8$

$R = 225 \text{ N}$

Resolving vertically

Correct equation

M1

A1

AG; correct force from correct working

A1

3

(d) $32.1 = 225\mu$

use of $F = \mu R$

M1

$\mu = \frac{32.1}{225} = 0.143$

Correct μ

A1

2

[9]

Q23.

(a) (i) $30 \sin 60^\circ t - 4.9t^2 = 0$

$$t = \frac{30 \sin 60^\circ}{4.9} = 5.302 \quad \text{AG}$$

Expression for height equal to zero

M1

Correct expression

A1

AG; correct conclusion from correct working

A1

3

(ii) $s = 30 \cos 60^\circ \times 5.302 = 79.5 \text{ m}$

Use of horizontal component of velocity to find the range

M1

Correct range

A1

2

(b) $30 \sin 60^\circ t - 4.9t^2 = 5$

$$4.9t^2 - 30 \sin 60^\circ t + 5 = 0$$

$$t = 0.200 \text{ or } 5.102$$

$$s = 30 \cos 60^\circ \times 5.102 = 76.5 \text{ m}$$

Expression for height equal to 5

M1

Correct LHS

A1

Correct RHS

A1

Solving a quadratic equation

m1

Correct solutions

A1

Correct distance

A1

6

[11]

Q24.

- (a) Ball is a particle

One appropriate assumption

B1

No air resistance

Second appropriate assumption

B1

2

- (b) (i)

$$0 = 12 \sin 40^\circ - 9.8t$$

$$t = \frac{12 \sin 40^\circ}{9.8} = 0.787 \text{ s}$$

Equation to find time at maximum height

M1

Correct equation

A1

Solving for t

M1

Correct time

A1

4

(ii) $h = 12 \sin 40^\circ \times 0.7871 - 4.9 \times 0.7871^2$
 $= 3.04 \text{ m}$

Substituting time from previous into expression for height

M1

Correct expression

A1

AG; correct height from correct working

A1

3

(c) $2.44 = 12 \sin 40^\circ t - 4.9t^2$

$$4.9t^2 - 12 \sin 40^\circ t + 2.44 = 0$$

$$t = 0.4385 \text{ or } 1.136$$

$$s = 12 \cos 40^\circ \times 1.136 = 10.4 \text{ m}$$

*Equation for time to get to the bar,
based on height being 2.44*

Correct LHS

M1

Correct RHS

A1

Solving quadratic

A1

Correct time / times

m1

*Substituting their larger time into an
expression for the horizontal displacement*

A1

Correct distance

M1

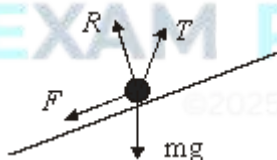
A1

7

[16]

Q25.

(a)



B1: Correct force diagram

B1

1

(b) $R + 20 \sin 30^\circ = 6g \cos 10^\circ$

*M1: Resolving perpendicular to the
slope with 3 terms*

A1: Correct equation

M1A1

$$R = 6g \cos 10^\circ - 20 \sin 30^\circ$$

dM1: Solving for R

dM1

$$R = 47.9 \text{ N (to 3 sf)} \quad \text{AG}$$

A1: Correct R from correct working

A1

4

(c) $F = \mu R$

M1: Use of $F = \mu R$

M1

$$6 \times 0.4 = 20 \cos 30^\circ - 6g \sin 10^\circ - \mu R$$

M1: Resolving parallel to slope to get 4 term equation of motion

A1: Correct equation

M1A1

$$\mu R = 4.710$$

A1: Correct $F / \mu R$

A1

$$\mu = \frac{4.710}{47.91} = 0.0983$$

dM1: Solving for μ

dM1

A1: AWR 0.098

A1

6

[11]

Q26.

(a) $x = 40t \cos \alpha$

M1

$$y = 40t \sin \alpha - \frac{1}{2}gt^2$$

M1A1

$$t = \frac{x}{40 \cos \alpha}$$

$$y = 40 \left(\frac{x}{40 \cos \alpha} \right) \sin \alpha - \frac{1}{2} g \left(\frac{x}{40 \cos \alpha} \right)^2$$

m1

$$y = x \tan \alpha - \frac{gx^2}{3200 \cos^2 \alpha}$$

$$y = x \tan \alpha - \frac{gx^2}{3200 (1 + \tan^2 \alpha)}$$

Answer given

A1

5

$$(b) \quad 4 = 100 \tan \alpha - \frac{9.8 \times 100^2}{3200 (1 + \tan^2 \alpha)}$$

M1

$$245 \tan^2 \alpha - 800 \tan \alpha + 277 = 0$$

M1A1

$$\tan \alpha = \frac{400 \pm \sqrt{(-400)^2 - (245)(277)}}{245}$$

Or equivalent

A1F

M1

$\alpha = 71^\circ, 21^\circ$ (22° acceptable)
(or 1.24 rad., 0.375 rad.)

A.T.W.R.T

(or equivalent in radians)

Must be positive

A1F

6

(c) The ball is a particle

E1

No air resistance, etc

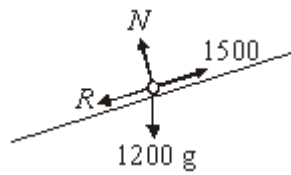
E1

2

[13]

Q27.

(a) (i)



3 correct & labelled

B1

All correct & labelled, no extras

B1

2

Ignore additional components of weight

(ii) $1500 = R + 1200 \times 9.8 \times \frac{1}{14}$

*M1: 3 terms with component attempted
($\alpha = 4.096$) – 1 each error*

M1A2

$R = 660$
AWRT

A1

4

(iii) 15 ms^{-1}

B1

1

(b) (i) $\frac{1500}{-1200}$

Accept $\pm$

M1

Mag of retardation = 1.25 ms^{-2}

A1

2

(ii) $0 = 15 + (-1.25) t$

velocities correct, but accept $\pm$ acceleration

M1

$t = 27.3 \text{ sec} \quad (27.27)$

ft one slip

A1F

2

[11]

Q28.

(a) $0 = (14 \cos 60)^2 - 2 \times 9.8 \times h$

Full method and component of u attempted, v may be present

M1

-1 each error including $v \neq 0$

A2

$h = 2.5 \text{ m}$

AWRT $ft \cos 30$, or $+g$ used

A1F

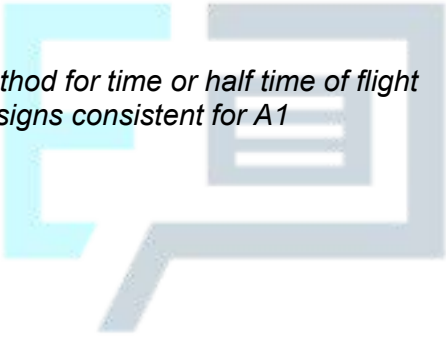
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(b) $2.5 = \frac{1}{2} \times 9.8 \times t^2$

*M1: full method for time or half time of flight
ft h , need signs consistent for A1*

M1A1F

$t = 0.714 \text{ sec} \left(\text{or } \frac{5}{7} \right)$


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A1F

3

[7]

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Q29.

(a) $5 = 10 \cos \theta^\circ$

Component attempted

M1

$\theta = 60$

AG

A1

2

(b) $Q = 10 \sin \theta$

Component attempted

M1

$$Q = 5\sqrt{3} \text{ N} = 8.66\text{N}$$

Accept either form

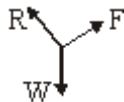
A1

2

[4]

Q30.

(a)



Three forces labelled & with arrows, W, mg or 35 g vertical (or 2 components of W), R and F perpendicular (ignore pairs of components of existing forces)

B1

1

(b) $R = 35g \times \cos 25^\circ$

Component attempted & g present for M1

M1A1

$$R = 311 \text{ N}$$

(R = 310.86) accept AWRT 311

A1F

3

(c) $F = 35g \times \sin 25^\circ$

Component attempted & g present for M1 & acceleration zero

M1A1

$$(= 144.96)$$

$$F = \mu R, 144.96 = \mu \times 310.86$$

Use of friction law with candidate's values, must have tried to find F

M1

$$\mu = 0.466$$

ft R and F, provided $\mu > 0$

M1A0 if $F < \mu R$ used

SC accept use of $\mu = \tan \theta$

A1F

4

[8]

Q31.

(a) $1200 - R = 1000 \times 0.25$

M1: all relevant terms used

M1A1

$R = 950 \text{ N}$

f.t. one slip if $R > 0$

A1F

3

(b) $2100 - 1000g \times 0.1 - 950 = 1000 \times a$

M1: 3 or 4 terms considered

M1

*-1 each term incorrect (any error),
or missing, $\alpha = 5.74^\circ$*

A1A1

$a = 0.17 \text{ ms}^{-2}$

f.t. one error if 4 terms considered

A1F

4

[7]

Q32.

(a) Sum of vertical forces = zero

used

M1

$4\cos 60^\circ + 8\cos 60^\circ - q = 0$

M1 for equation, 3 forces

M1 A1

$q = 6$

cao

A1

4

(b) horizontal:

$$R = 3 + 8 \cos 30^\circ - 4 \cos 30^\circ$$

M1A1

$$R = 6.46 \text{ or } 6.47$$

A1F

3

[7]

Q33.

(a) $6 = a \cos 60^\circ + 8 \cos 60^\circ$

3 terms & attempt at components for M1 correct components

M1

$$6 = \frac{a}{2} + 4$$

signs correct

A1A1

$$a = 4$$

A1

4

(b) $\pm(8 \cos 30^\circ - a \cos 30^\circ)$

*Components parallel to Oy only for M1
signs correct, 8 and a (or 4) used*

M1A1

$$3.46 \text{ or } 3.47$$

ft one error e.g. sign or incorrect a

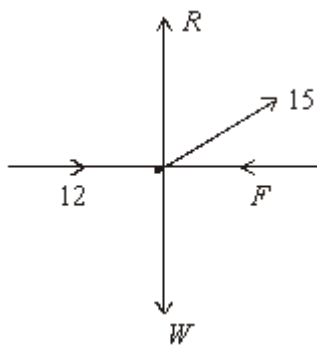
A1F

3

[7]

Q34.

(a)



any 3 correct, including arrows and labels

B1

all correct, including arrows and labels

B1

2

accept mg for W

accept either 15 N **or** components, but **not** both

(b) $F = 12 + 15 \cos 30^\circ$

for M1, both terms on RHS and sin/cos

M1A1

$= 12 + 12.99$

$F = 25 \text{ N}$

CAO

A1

3

(c) $25 = \frac{1}{3} R$

M1A0 if inequality used

M1

$R = 75 \text{ N}$

AWRT; ft F ; accept use of c's F from (b);

accept 0.333 for μ

A1F

2

(b) $R + 15 \cos 60^\circ = W$

3 terms and attempt at component

$Wg \Rightarrow$ M1A0A0

M1

$$75 + 7 \frac{1}{2} = W$$

ft R

A1F

$$W = 82 \frac{1}{2} \text{ N}$$

AWRT 82.5, ft R or one slip

A1F

3

[10]

Q35.

(a) $0 = 21 \times 0.7 - 9.8t$

*Write 'wrong g' at first use of 100 or 9.81 and award A0.
complete method with attempt at
component; accept $\alpha = 44.4(3)$; accept $\pm g$*

M1 A1

$t = 1.5$

*If $\alpha = 44.4$ used, or 2 stage method used,
accept values of t which round to 1.5;
($g = 10$ gives 1.47)*

A1

3

(b) (i) $S = 21 \times 0.7 \times 1.5 - \frac{1}{2} \times 9.8 \times 1.5^2$

*complete method with attempt at
component; accept c's time and $\pm g$*

M1

or $0 = (21 \times 0.7)^2 - 2 \times 9.8 \times S$

all correct, (accept c's time) including $-g$ and $\sin \alpha$

A1

$S = 11.025$

Accept c's time; ft one slip e.g. $+g$ or $\cos \alpha$

A1F

Distance above tree = $11.025 - 8 \approx 3\text{m}$

ft if $d > 0$,

$g = 10$ gives $S = 10.8$ and $d = 2.8$ (accept)

A1F

4

(ii) $S = 8$

$$8 = 21 \times 0.7t - \frac{1}{2} \times 9.8 \times t^2$$

*full method for t or v with attempt at component ;
accept $\pm g$*

all correct, including $-g$ and $\sin \alpha$,

but condone repeated error in $-g$ and $\sin \alpha$ from (i)

M1 A1

$$4.9t^2 - 14.7t + 8 = 0$$

$$t = \frac{14.7 \pm \sqrt{14.7^2 - 4 \times 4.9 \times 8}}{2 \times 4.9}$$

*To find t by solving equation or using $v (=7.7)$
in $v = u + at$*

Substituting; ft

m1 A1F

$$t = 0.71 \text{ (0.7143)}$$

CAO

A1

5

(iii) time = $2.29 - 0.71$

*Subtraction of two positive values of t ,
or use of symmetry*

M1

$$\text{or } 2(1.5 - 0.71)$$

Allow values between 2.28, 2.29 and 0.71, 0.72

$$= 1.57 \text{ sec}$$

Accept 1.56 – 1.58 ($g = 10$ gives $t = 1.5$)

A1F

2

Alternative: If $\cos 44.4$ used:

(b)(i) $S = 11.475$, $d = 3.475$

(ii) $t = 0.688$ (2.373)

(iii) $t = 1.69$

If no component:

(a) M0;

(b)(i) SC: $d = 12.5$

M1 A1 for complete method

(ii) SC: $t = 0.423$

M1 A1 for complete method

(iii) $t = 3.42$

M1A1

[14]



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